Speculations Concerning the First Ultraintelligent Machine*

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1. Introduction

The survival of man depends on the early construction of an ultra-intelligent machine. In order to design an ultraintelligent machine we need to understand more about the human brain or human thought or both. In the following pages an attempt is made to take more of the magic out of the brain by means of a "subassembly" theory, which is a modification of Hebb's famous speculative cell-assembly theory. My belief is that the first ultraintelligent machine is most likely to incorporate vast artificial neural circuitry, and that its behavior will be partly explicable in terms of the subassembly theory. Later machines will all be designed by ultra- * Based on talks given in a Conference on the Conceptual Aspects of

Biocommunications, Neuropsychiatric Institute, University of California, Los Angeles, October 1962; and in the Artificial Intelligence Sessions of the Winter General Meetings of the IEEE, January 1963 [1, 46]. The first draft of this monograph was completed in April 1963, and the present slightly amended version in May 1964. I am much indebted to Mrs. Euthic Anthony of IDA for the arduous task of typing.

intelligent machines, and who am I to guess what principles they will devise? But probably Man will construct the deus ex machina in his own image. The subassembly theory sheds light on the physical embodiment of memory and meaning, and there can be little doubt that both will need embodiment in an ultraintelligent machine. Even for the brain, we shall argue that physical embodiment of meaning must have originated for reasons of economy, at least if the metaphysical reasons can be ignored. Economy is important in any engineering venture, but especially so when the price is exceedingly high, as it most likely will be for the first ultraintelligent machine. Hence semantics is relevant to the design of such a machine. Yet a detailed knowledge of semantics might not be required, since the artificial neural network will largely take care of it, provided that the parameters are correctly chosen, and provided that the network is adequately integrated with its sensorium and motorium (input and output). For, if these conditions are met, the machine will be able to learn from experience, by means of positive and negative reinforcement, and the instruction of the machine will resemble that of a child. Hence it will be useful if the instructor knows something about semantics, but not necessarily more useful than for the instructor of a child. The correct choice of the parameters, and even of the design philosophy, will depend on the usual scientific method of successive approximation, using speculation, theory, and experiment. The percentage of speculation needs to be highest in the early stages of any endeavor. Therefore no apology is offered for the speculative nature of the present work. For we are certainly still in the early stages in the design of an ultraintelligent machine. In order that the arguments should be reasonably self-contained, it is necessary to discuss a variety of topics. We shall define an ultraintelligent machine, and, since its cost will be very large, briefly consider its potential value. We say something about the physical embodiment of a word or statement, and defend the idea that the function of meaning is economy by describing it as a process of ”regeneration.” In order to explain what this means, we devote a few pages to the nature of communication. (The brain is of course a complex communication and control system.) We shall need to discuss the process of recall, partly because its understanding is very closely related to the understanding of understanding. The process of recall in its turn is a special case of statistical information retrieval. This subject will be discussed in Section 5. One of the main difficulties in this subject is how to estimate the probabilities of events that have never occurred. That such probabilities are relevant to intelligence is to be expected, since intelligence is sometimes defined as the ability to adapt to new circumstances.

The difficulty of estimating probabilities is sometimes overlooked in the literature of artificial intelligence, but this article would be too long if the subject were surveyed here. A separate monograph has been written on this subject [48]. Some of the ideas of Section 5 are adapted, in Section 6, to the problem of recall, which is discussed and to some extent explained in terms of the subassembly theory. The paper concludes with some brief suggestions concerning the physical representation of ”meaning.” This paper will, as we said, be speculative: no blueprint will be suggested for the construction of an ultraintelligent machine, and there will be no reference to transisitors, diodes, and cryogenics. (Note, however, that cryogenics have the important merit of low power consumption. This feature will be valuable in an ultraintelligent machine.) One of our aims is to pinpoint some of the difficulties. The machine will not be on the drawing board until many people have talked big, and others have built small, conceivably using deoxyribonucleic acid (DNA). Throughout the paper there are suggestions for new research. Some further summarizing remarks are to be found in the Conclusions.

2. Ultraintelligent Machines and Their Value

Let an ultraintelligent machine be defined as a machine that can far surpass all the intellectual activities of any man however clever. Since the design of machines is one of these intellectual activities, an ultraintelligent machine could design even better machines; there would then unquestionably be an ”intelligence explosion,” and the intelligence of man would be left far behind (see for example refs. [22], [34], [44]). Thus the first ultraintelligent machine is the last invention that man need ever make, provided that the machine is docile enough to tell us how to keep it under control. It is curious that this point is made so seldom outside of science fiction. It is sometimes worthwhile to take science fiction seriously. In one science fiction story a machine refused to design a better one since it did not wish to be put out of a job. This would not be an insuperable difficulty, even if machines can be egotistical, since the machine could gradually improve itself out of all recognition, by acquiring new equipment. B. V. Bowden stated on British television (August 1962) that there is no point in building a machine with the intelligence of a man, since it is easier to construct human brains by the usual method. A similar point was made by a speaker during the meetings reported in a recent IEEE

publication [7], but I do not know whether this point appeared in the published report. This shows that highly intelligent people can overlook the ”intelligence explosion.” It is true that it would be uneconomical to build a machine capable only of ordinary intellectual attainments, but it seems fairly probable that if this could be done then, at double the cost, the machine could exhibit ultraintelligence. Since we are concerned with the economical construction of an ultraintelligent machine, it is necessary to consider first what such a machine would be worth. Carter [11] estimated the value, to the world, of J. M. Keynes, as at least 100,000 million pounds sterling. By definition, an ultraintelligent machine is worth far more, although the sign is uncertain, but since it will give the human race a good chance of surviving indefinitely, it might not be extravagant to put the value at a megakeynes. There is the opposite possibility, that the human race will become redundant, and there are other ethical problems, such as whether a machine could feel pain especially if it contains chemical artificial neurons, and whether an ultraintelligent machine should be dismantled when it becomes obsolete [43, 84]. The machines will create social problems, but they might also be able to solve them in addition to those that have been created by microbes and men. Such machines will be feared and respected, and perhaps even loved. These remarks might appear fanciful to some readers, but to the writer they seem very real and urgent, and worthy of emphasis outside of science fiction. If we could raise say a hundred billion dollars we might be able to simulate all the neurons of a brain, and of a whole man, at a cost of ten dollars per artificial neuron. But it seems unlikely that more than say a millikeynes would actually be forthcoming, and even this amount might be difficult to obtain without first building the machine! It would be justified if, with this expenditure, the chance of success were about 10^{-9}. Until an ultraintelligent machine is built perhaps the best intellectual feats will be performed by men and machines in very close, sometimes called "symbiotic," relationship, although the term "biomechanical" would be more appropriate. As M. H. A. Newman said in a private communication in 1946, an electronic computer might be used as "rough paper" by a mathematician. It could already be used in this manner by a chess player quite effectively, although the effectiveness would be much increased if the chess-playing programs were written with extremely close man-machine interaction in mind from the start. The reason for this effectiveness is that the machine has the advantage in speed and accuracy for routine calculation, and man has the advantage in imagination. Moreover, a large part of imagination in chess can be reduced to routine. Many of the ideas that require imagination in the amateur are routine for the master. Consequently the machine might

appear imaginative to many observers and even to the programmer. Similar comments apply to other thought processes. The justification for chess-playing programs is that they shed light on the problem of artificial intelligence without being too difficult to write. Their interest would be increased if chess were replaced by so-called "randomized chess," in which the positions of the white pieces on the first rank are permuted at random before the game begins (but with the two bishops on squares of opposite colors), and then the initial positions of the black pieces are determined by mirror symmetry. This gives rise to 1440 essentially distinct initial positions and effectively removes from the game the effect of mere parrot learning of the openings, while not changing any of the general principles of chess. In ordinary chess the machine would sometimes beat an international Grandmaster merely by means of a stored opening trap, and this would be a hollow victory. Furthermore a program for randomized chess would have the advantage that it would not be necessary to store a great number of opening variations on magnetic tape. The feats performed by very close man-machine interaction by say 1980 are likely to encourage the donation of large grants for further development. By that time, there will have been great advances in microminiaturization, and pulse repetition frequencies of one billion pulses per second will surely have been attained in large computers (for example see Shoulders [91]). On the other hand, the cerebral cortex of a man has about five billion neurons, each with between about twenty and eighty dendrites ([90], pp. 35 and 51), and thousands of synapses. (At the recent IEEE meetings, P. Mueller offered the estimate 300,000 orally. It would be very interesting to know the corresponding figure for the brain of a whale, which, according to Tower [99], has about three times as many neurons as a human brain. Perhaps some whales are ultraintelligent! [49].) Moreover, the brain is a parallel-working device to an extent out of all proportion to any existing computer. Although computers are likely to attain a pulse repetition speed advantage of say a million over the brain, it seems fairly probable, on the basis of this quantitative argument, that an ultraintelligent machine will need to be ultraparallel. In order to achieve the requisite degree of ultraparallel working it might be useful for many of the elements of the machine to contain a very short-range microminiature radio transmitter and receiver. The range should be small compared with the dimensions of the whole machine. A "connection" between two close artificial neurons could be made by having their transmitter and receiver on the same or close frequencies. The strength of the connection could be represented by the accuracy of the tuning. The receivers would need numerous filters so as to be

35

able to distinguish a large number of different frequencies. It might be better still to use laser beams in a similar manner. The problems of embodiment of an ultraintelligent machine are not discussed further in this paper.

3. Communication as Regeneration

In order to understand the brain, it is necessary to understand the nature of communication. A communication system consists of a source, a channel, and a destination. The source generates a message, which is transmitted through the channel to the destination. The message is a sequence of symbols, such as letters or words. The channel is the medium through which the message is transmitted, such as a wire or the air. The destination is the person or machine that receives the message. A communication system is said to be "noisy" if the message is distorted during transmission. The distortion is called "noise." The problem of communication in the presence of noise is to design a system that can transmit messages with a small probability of error. The theory of communication was founded by C. E. Shannon in 1948 [41]. Shannon's theory is a mathematical theory of communication that is concerned with the problem of transmitting information through a noisy channel. The theory is based on the concept of "information," which is a measure of the uncertainty of an event. The amount of information in a message is a measure of the reduction of uncertainty that is produced by the message. Shannon's theory has been very successful in the design of communication systems. It has also had a profound influence on other fields, such as psychology, linguistics, and computer science. The brain can be regarded as a communication system. The sense organs are the source, the nervous system is the channel, and the brain is the destination. The messages are the nerve impulses that are transmitted from the sense organs to the brain. The brain processes these messages and produces a response. The brain is a very complex communication system. It is much more complex than any man-made communication system. The brain is able to learn from experience, and it is able to adapt to new situations. These are two of the most important characteristics of an intelligent system. The process of communication can be regarded as a process of "regeneration." The source generates a message, which is a representation of some "meaning." The message is transmitted through the channel to the destination. The destination receives the message and regenerates the meaning. The meaning is not transmitted directly from the source to the destination. It is regenerated at the destination. The process of regeneration is very important in the brain. The brain is constantly regenerating meanings from the messages that it receives from the sense organs. This is how the brain is able to understand the world. The process of regeneration is also very important in an ultraintelligent machine. The machine will need to be able to regenerate meanings from the messages that it receives from its environment. This is how the machine will be able to understand the world. The process of regeneration is a form of "pattern recognition." The destination recognizes the pattern in the message and regenerates the meaning. The pattern is a representation of the meaning. The problem of pattern recognition is one of the most important problems in the field of artificial intelligence. A great deal of research has been done on this problem, but it is still not well understood. The subassembly theory, which is described in Section 6, is a theory of how the brain recognizes patterns. The theory is based on the idea that the brain is a network of "cell assemblies." A cell assembly is a group of neurons that are strongly connected to each other. When a pattern is presented to the brain, a cell assembly is activated. The activation of the cell assembly is a representation of the pattern.

36

THE FIRST ULTRAINTELLIGENT MACHINE

The subassembly theory is a speculative theory, but it is consistent with what is known about the brain. The theory has been used to explain a number of psychological phenomena, such as perception, memory, and learning. The subassembly theory can be used to design an ultraintelligent machine. The machine would be a network of artificial neurons. The neurons would be connected to each other in such a way that they form cell assemblies. The machine would be able to learn from experience by modifying the connections between the neurons. The machine would be able to recognize patterns by activating cell assemblies. The activation of a cell assembly would be a representation of the pattern. The machine would be able to regenerate meanings by activating cell assemblies. The activation of a cell assembly would be a representation of the meaning. The machine would be able to understand the world by regenerating meanings from the messages that it receives from its environment. The machine would be a truly intelligent machine.

4. Some Representations of "Meaning" and Their Relevance to Intelligent Machines

The meaning of a word or a statement is a very difficult concept to define. The meaning of a word is not the same as the word itself. The meaning of a statement is not the same as the statement itself. The meaning of a word or a statement is something that is in the mind of the person who uses the word or the statement. The meaning of a word can be represented in a number of ways. One way is to use a dictionary. A dictionary is a book that contains a list of words and their meanings. The meaning of a word is given by a definition. A definition is a statement that explains the meaning of a word. Another way to represent the meaning of a word is to use a thesaurus. A thesaurus is a book that contains a list of words and their synonyms. A synonym is a word that has the same or nearly the same meaning as another word. A third way to represent the meaning of a word is to use a semantic network. A semantic network is a graph in which the nodes represent concepts and the edges represent the relationships between the concepts. The meaning of a word is represented by a node in the network. The relationships between the meanings of words are represented by the edges in the network. The meaning of a statement can also be represented in a number of ways. One way is to use a logical formula. A logical formula is a statement that is written in a formal language. The meaning of the statement is given by the truth value of the formula. The truth value of a formula is either true or false. Another way to represent the meaning of a statement is to use a semantic network. The meaning of the statement is represented by a subgraph of the network. The subgraph represents the relationships between the concepts in the statement.

The problem of representing meaning is a very important problem in the field of artificial intelligence. An ultraintelligent machine will need to be able to represent meanings in order to understand the world. The subassembly theory provides a way to represent meanings in the brain. The meaning of a word or a statement is represented by a cell assembly. The activation of the cell assembly is a representation of the meaning. The subassembly theory can be used to design an ultraintelligent machine that can represent meanings. The machine would be a network of artificial neurons. The neurons would be connected to each other in such a way that they form cell assemblies. The machine would be able to learn from experience by modifying the connections between the neurons. The machine would be able to represent meanings by activating cell assemblies. The activation of a cell assembly would be a representation of the meaning. The machine would be able to understand the world by representing the meanings of the messages that it receives from its environment. The machine would be a truly intelligent machine.

5. Recall and Information Retrieval

Recall is the process of retrieving information from memory. Information retrieval is the process of finding documents that are relevant to a query. Recall and information retrieval are two of the most important problems in the field of artificial intelligence. The problem of recall can be stated as follows: given a cue, retrieve from memory the information that is associated with the cue. The cue can be a word, a phrase, a question, or an image. The information that is retrieved can be a word, a phrase, a statement, or an image. The problem of information retrieval can be stated as follows: given a query, find the documents that are relevant to the query. The query can be a word, a phrase, or a question. The documents can be text documents, images, or videos. The subassembly theory provides a way to solve the problems of recall and information retrieval. The theory is based on the idea that the brain is a network of cell assemblies.

A cell assembly is a group of neurons that are strongly connected to each other. When a cue is presented to the brain, a cell assembly is activated. The activation of the cell assembly is a representation of the cue. The activation of the cell assembly spreads to other cell assemblies that are associated with the cue. The activation of these other cell assemblies is a representation of the information that is retrieved from memory. The subassembly theory can be used to design an ultraintelligent machine that can solve the problems of recall and information retrieval. The machine would be a network of artificial neurons. The neurons would be connected to each other in such a way that they form cell assemblies. The machine would be able to learn from experience by modifying the connections between the neurons.

38

THE FIRST ULTRAINTELLIGENT MACHINE

The machine would be able to solve the problem of recall by activating cell assemblies. When a cue is presented to the machine, a cell assembly is activated. The activation of the cell assembly spreads to other cell assemblies that are associated with the cue. The activation of these other cell assemblies is a representation of the information that is retrieved from memory. The machine would be able to solve the problem of information retrieval by activating cell assemblies. When a query is presented to the machine, a cell assembly is activated. The activation of the cell assembly spreads to other cell assemblies that are associated with the query. The activation of these other cell assemblies is a representation of the documents that are relevant to the query. The machine would be a truly intelligent machine.

6. Cell Assemblies and Subassemblies

The subassembly theory is a modification of Hebb's cell-assembly theory [20]. Hebb's theory is a speculative theory of how the brain works. The theory is based on the idea that the brain is a network of neurons. The neurons are connected to each other by synapses. The strength of a synapse can be modified by experience. Hebb's theory can be summarized as follows:

1. When two neurons are active at the same time, the synapse between them is strengthened.
2. A cell assembly is a group of neurons that are strongly connected to each other.
3. When a part of a cell assembly is activated, the whole cell assembly is activated.
4. The activation of a cell assembly is a representation of a concept.

The subassembly theory is a modification of Hebb's theory. The subassembly theory can be summarized as follows:

1. A cell assembly is a group of neurons that are strongly connected to each other.
2. A subassembly is a subset of a cell assembly.
3. When a part of a subassembly is activated, the whole subassembly is activated.
4. The activation of a subassembly is a representation of a concept.
5. The meaning of a concept is represented by the set of subassemblies that are activated when the concept is presented to the brain.

The subassembly theory is a more powerful theory than Hebb's theory. The subassembly theory can explain a number of psychological phenomena that cannot be explained by Hebb's theory. For example, the subassembly theory can explain how the brain can represent the meaning of a word. The meaning of a word is represented by the set of subassemblies that are activated when the word is presented to the brain. Each subassembly represents a different aspect of the meaning of the word. The subassembly theory can also explain how the brain can learn new concepts. A new concept is learned by creating a new subassembly. The new subassembly is created by combining existing subassemblies.

39

The subassembly theory can be used to design an ultraintelligent machine. The machine would be a network of artificial neurons. The neurons would be connected to each other in such a way that they form cell assemblies and subassemblies. The machine would be able to learn from experience by modifying the connections between the neurons. The machine would be able to represent concepts by activating subassemblies. The meaning of a concept would be represented by the set of subassemblies that are activated when the concept is presented to the machine. The machine would be able to learn new concepts by creating new subassemblies. The new subassemblies would be created by combining existing subassemblies. The machine would be a truly intelligent machine.

7. An Assembly Theory of Meaning

The subassembly theory provides a way to represent the meaning of a word or a statement. The meaning of a word or a statement is represented by a set of subassemblies. Each subassembly represents a different aspect of the meaning. For example, the meaning of the word ”dog” can be represented by a set of subassemblies. One subassembly might represent the visual appearance of a dog. Another subassembly might represent the sound of a dog barking. A third subassembly might represent the fact that a dog is a mammal. The meaning of the statement ”The dog is barking” can be represented by a set of subassemblies. One subassembly might represent the meaning of the word ”dog.” Another subassembly might represent the meaning of the word ”barking.” A third subassembly might represent the relationship between the dog and the barking. The subassembly theory can be used to design an ultraintelligent machine that can understand the meaning of words and statements. The machine would be a network of artificial neurons. The neurons would be connected to each other in such a way that they form cell assemblies and subassemblies. The machine would be able to learn from experience by modifying the connections between the neurons. The machine would be able to understand the meaning of a word or a statement by activating a set of subassemblies. The set of subassemblies would represent the meaning of the word or the statement. The machine would be a truly intelligent machine.

8. The Economy of Meaning

The subassembly theory provides a way to represent the meaning of a word or a statement in an economical way. The meaning of a word or a statement is represented by a set of subassemblies. Each subassembly can be used to represent a different aspect of the meaning. This means that the same subassembly can be used in the representation of the meaning of many different words and statements. For example, the subassembly that represents the visual appearance of a dog can be used in the representation of the meaning of the word ”dog,” the word ”poodle,” and the word ”beagle.” The subassembly that represents the sound of a dog barking can be used in the representation of the meaning of the statement ”The dog is barking” and the statement ”I hear a dog barking.”

The use of subassemblies to represent meaning is a very economical way to store information. The brain uses this method to store information, and an ultraintelligent machine could also use this method.

9. Conclusions

The survival of man depends on the early construction of an ultraintelligent machine. An ultraintelligent machine is a machine that can far surpass all the intellectual activities of any man however clever. The first ultraintelligent machine is the last invention that man need ever make. The subassembly theory is a speculative theory of how the brain works. The theory can be used to design an ultraintelligent machine. The machine would be a network of artificial neurons. The neurons would be connected to each other in such a way that they form cell assemblies and subassemblies. The machine would be able to learn from experience by modifying the connections between the neurons. The machine would be able to understand the world by regenerating meanings from the messages that it receives from its environment. The machine would be able to represent meanings by activating subassemblies. The machine would be able to learn new concepts by creating new subassemblies. The machine would be a truly intelligent machine.

10. Appendix: Informational and Causal Interactions

The distinction between informational and causal interactions is a very important one. An informational interaction is an interaction in which information is transferred from one system to another. A causal interaction is an interaction in which one system causes a change in another system. For example, when you read a book, there is an informational interaction between you and the book. Information is transferred from the book to you. When you throw a ball, there is a causal interaction between you and the ball. You cause the ball to move. The distinction between informational and causal interactions is not always clear. For example, when you speak to someone, there is both an informational and a causal interaction. You transfer information to the other person, and you also cause the other person to have a certain mental state. The subassembly theory can be used to explain the distinction between informational and causal interactions. An informational interaction is an interaction in which a subassembly is activated. A causal interaction is an interaction in which a subassembly is created or modified.

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